

Interpretability-Guided Layer Pruning for Efficient Multilingual Machine Translation

V. Adrian Castillo

Brigham Young University
vac12@byu.edu

Stephen Richardson

Brigham Young University
srichardson@byu.edu

Abstract

Multilingual large language models (LLMs) have demonstrated exceptional performance in machine translation tasks, yet their significant computational requirements pose substantial barriers to real-world deployment in resource-constrained environments. Model compression techniques such as quantization and layer pruning have emerged as promising solutions, but existing iterative pruning approaches suffer from quality degradation and offer limited insight into why certain layers can be safely removed. In this work, we investigate whether interpretability techniques can identify which layers contribute least to translation quality in Aya Expanse 8B [Dang et al., 2024], enabling more principled pruning decisions. We apply Information Flow Routes [Ferrando and Voita, 2024] to analyze layer importance, apply LoRA fine-tuning and Knowledge Distillation for quality recovery, and combine the resulting pruned models with GPTQ quantization [Frantar et al., 2023]. Our experiments on Czech→German and English→Spanish show that IFR-guided pruning matches or outperforms iterative pruning at moderate compression levels at a fraction of the compute cost, and that GPTQ quantization applied after recovery produces models that are one third the size and more than $2\times$ faster with minimal quality loss. These findings suggest that interpretability-guided pruning combined with static quantization is a practical and principled pathway toward deployment-ready compression of multilingual LLMs.

1 Introduction

Large Language Models (LLMs) have shown exceptional performance across natural language tasks. Studies have shown that multilingual LLMs are among the top choices for machine translation tasks, due to their strong cross-lingual capabilities [Kocmi et al., 2025]. These models typically contain billions of parameters and are trained on

massive amounts of text data. Their exceptional capabilities come with significant computational requirements, such as substantial memory and storage demands. Our subject of study is the Aya Expanse 8B model [Dang et al., 2024], which is a relatively small model and yet requires approximately 32 GB of GPU memory at full FP16 precision, needing high-end hardware such as an NVIDIA A100 GPU.

Such hardware demands pose significant challenges for real-world deployment, particularly in resource-constrained environments such as mobile devices, embedded systems, and edge computing platforms. This challenge is especially acute in developing regions, where internet access relies heavily on mobile devices and high-end computing infrastructure is scarce, limiting the adoption of such technologies. Compressing these models is therefore not merely a technical pursuit, it is a strategic priority with direct implications for global accessibility and the democratization of AI.

The three main approaches to model compression are quantization, layer pruning, and knowledge distillation [Zhu et al., 2023, Hinton et al., 2015]. Among these, quantization has emerged as the most popular approach for its simplicity and effectiveness, reducing the Aya Expanse 8B model size by up to 65% while nearly matching baseline translation quality [Gaido et al., 2025]. Layer pruning, typically identified through a greedy, iterative search process [Moslem et al., 2025], achieves meaningful size reduction but comes with notable degradation in translation quality, with COMET scores dropping from 55.3 to as low as 21.4 in the most aggressively pruned configurations [Gaido et al., 2025].

Rather than relying on such iterative methods, we propose an interpretability-guided method using Information Flow Routes (IFR) [Ferrando and Voita, 2024] to identify each layer’s contribution to translation quality before pruning, followed by

GPTQ quantization [Frantar et al., 2023] to produce a deployment-ready model compatible with optimized inference engines. Thus, we raise the following research questions:

- Does interpretability-guided pruning outperform iterative pruning at equivalent layer pruning rates?
- Does the resulting pruned model remain robust to subsequent quantization?

Answering these questions could open the door to more principled, robust compression pipelines and bring multilingual LLMs like Aya Expanse closer to deployment in resource-constrained, globally accessible settings.

2 Related Work

Model compression has become an active area of research as the deployment of large language models in resource-constrained settings grows increasingly important [Zhu et al., 2023]. The WMT 2025 Model Compression shared task [Gaido et al., 2025] represents the first systematic evaluation of compression techniques applied specifically to general-purpose LLMs for machine translation, using Aya Expanse 8B as the target model. The task revealed that quantization-based approaches were the dominant strategy among participants, with methods such as BitsAndBytes [Dettmers et al., 2022] and LeanAya achieving up to 65% size reduction while nearly preserving baseline translation quality. Although quality was preserved, inference speed was deeply impacted, which is critical for deployment.

Another strategy explored in the WMT 2025 shared task was layer pruning. Moslem et al. [2025] applied iterative layer pruning to Aya Expanse 8B for Czech→German and English→Egyptian Arabic translation, incrementally removing the layer with the least impact on translation quality at each step. Their primary submission reduced the model from 32 to 24 layers and from 8.03B to 6.28B parameters, with the authors reporting strong quality retention in their standalone paper. However, their results in the shared task evaluation [Gaido et al., 2025] show a more pronounced quality drop, with COMET scores falling from 55.3 to 39.9 for Czech→German, a notable discrepancy that raises questions about the robustness of the method across different evaluation conditions.

These limitations motivate more principled, interpretability-guided alternatives to layer selection. Ferrando and Voita [2024] propose Information Flow Routes, a method that traces how information propagates through a transformer by building attribution graphs over the residual stream, identifying which attention heads and feed-forward layers are most critical for a given prediction, including components specialized for multilingual text processing. Related attribution-guided pruning approaches [Hatefi et al., 2025] have also been explored for general-purpose LLM compression.

3 Data

We use the News Commentary v18 parallel corpora from the OPUS collection [Tiedemann, 2012] for Czech→German and English→Spanish. We chose these two language pairs to compare our results with Moslem et al. [2025] and extend their work to an additional language pair. We apply a filtering pipeline of four steps: deduplication, length filtering, language detection, and semantic similarity filtering. Length filtering removes segments longer than 200 words and those with a source/target length ratio above $1.5\times$. Language detection uses fastText lid.176.bin [Joulin et al., 2017] with a confidence threshold of ≥ 0.9 for both source and target. Semantic similarity filtering uses the paraphrase-multilingual-MiniLM-L12-v2 sentence-transformers model [Reimers and Gurevych, 2019] with a cosine similarity threshold of ≥ 0.7 , instead of the mUSE model used by Moslem et al. [2025]. After filtering, we split the data into a training set of 100,000 randomly sampled segments and a test set of 500 segments. Table 1 shows the corpus statistics at each filtering stage.

We also generate a synthetic dataset for the Knowledge Distillation [Hinton et al., 2015, Kim and Rush, 2016] experiments using the Aya Expanse 32B model as the teacher. We translate the training split of each language pair and keep only the translations with a COMET score above 0.7, giving 98,377 segments for Czech→German and 65,482 for English→Spanish.

Table 1: Corpus statistics at each filtering stage.

Stage	cs→de	en→es
Raw	243,807	499,368
Deduplication	241,831	498,569
Length Filter	225,680	464,721
Lang. Detection	214,532	374,216
Sem. Sim.	208,812	368,477
Train Split	100,000	100,000
Test Split	500	500
KD Data	98,377	65,482

4 Methodology

Our experiments are conducted on the Aya Expanse 8B model [Dang et al., 2024], an open-weight multilingual model with 32 decoder layers that supports 23 languages including Czech, German, English, and Spanish. This model is the compression target in all our experiments, consistent with the WMT 2025 Model Compression shared task [Gaido et al., 2025]. The Aya Expanse 32B model is used as the teacher model for Knowledge Distillation data generation, as described in Section 3.

Our pipeline has three stages: layer pruning, model recovery, and quantization. In the pruning stage we test a heuristic iterative baseline and an interpretability-guided approach at three compression levels. In the recovery stage we compare LoRA fine-tuning alone against LoRA combined with Knowledge Distillation. Finally, we apply quantization to test the robustness of the compressed models. All experiments are run for both Czech→German and English→Spanish.

4.1 Baseline Iterative Pruning

Given the discrepancy between the results of Moslem et al. [2025] in their standalone paper and in the WMT 2025 shared task, we start by running our own iterative layer pruning to have a solid baseline for comparison. We follow the same approach as Moslem et al. [2025]: at each step, we temporarily remove each candidate layer and evaluate the model on a 200-sentence validation set using chrF++ [Popović, 2017]. The first and last layers are always protected. The layer whose removal causes the smallest quality drop is permanently removed, and this process repeats until we reach the target number of layers. We prune for 8, 12, and 16 layers, giving models with 24, 20, and 16 layers respectively. Instead of full fine-tuning like Moslem et al. [2025], we use LoRA for recovery, as described in Section 4.3. We also run one full

fine-tuning variant to allow a direct comparison with their approach.

4.2 Interpretability-Guided Pruning

Rather than searching for the least important layers through iterative ablation, we use Information Flow Routes [Ferrando and Voita, 2024] to score each layer’s contribution to the model before pruning. For each layer, we register forward hooks to capture the layer input, attention output, MLP output, and layer output. The importance score is based on a proximity L1 measure:

$$\text{proximity}(y, s) = \text{clamp}(\|s\|_1 - \|s - y\|_1, 0) \quad (1)$$

We apply this to both the post-attention residual and the MLP layer output, normalizing so that the component and residual contributions sum to 1. The final score per layer is the sum of the attention and MLP importance, averaged across 200 translation examples. We then rank layers by score and select the least important ones for pruning.

We test two IFR-based pruning strategies. The first removes a fixed number of layers (8, 12, or 16) based on the ranking, directly comparable to the iterative baseline. The second is an adaptive approach where the number of layers removed is decided by the score distribution itself. We compute a threshold $\tau = \lambda \cdot \mu$, where μ is the mean importance score and λ is a threshold factor. Any layer scoring below τ is removed, except the first and last:

$$\text{remove} = \{i \mid s_i < \tau, 0 < i < L - 1\} \quad (2)$$

We test three threshold values $\lambda \in \{0.3, 0.5, 0.7\}$, which result in approximately 2, 9–10, and 14 layers removed on Aya Expanse 8B. The threshold approach is useful because it respects the shape of the score distribution rather than assuming a fixed number of removable layers.

4.3 Recovery Methods

After pruning, we apply LoRA fine-tuning [Hu et al., 2022] to recover translation quality. We apply adapters to all attention and MLP projection layers with rank $r=16$, $\alpha=32$, and dropout of 0.05. We train for 3 epochs with a learning rate of $2e-5$ and an effective batch size of 32. We compare three recovery setups: no recovery as a control, LoRA fine-tuning on the filtered authentic data,

and LoRA fine-tuning on a mix of authentic and sequence-level Knowledge Distillation [Kim and Rush, 2016] data (see Section 3).

4.4 Quantization

To test robustness to quantization, we apply GPTQ 4-bit quantization [Frantar et al., 2023] post-hoc to the recovered models using the `gptqmodel` library. GPTQ produces a serialized quantized model compatible with vLLM [Kwon et al., 2023], making it suitable for realistic deployment benchmarking. We also ran a small number of experiments with BitsAndBytes NF4 4-bit [Dettmers et al., 2023] and INT8 [Dettmers et al., 2022] quantization, but given that these are runtime quantization methods not compatible with vLLM, we do not focus on them in our analysis.

4.5 Evaluation

We evaluate all models using BLEU [Papineni et al., 2002], chrF++ [Popović, 2017], and COMET [Rei et al., 2020] on the 500-sentence test set. We compute BLEU and chrF++ using sacreBLEU [Post, 2018] to ensure reproducibility. For COMET we use the `Unbabel/wmt22-comet-da` model, consistent with Moslem et al. [2025]. Inference speed is measured separately using vLLM with 5 warmup prompts and 50 timed prompts at greedy decoding, reporting tokens per second for both fp16 and GPTQ-4bit variants.

5 Experimental Setup

Our experiments are designed to answer the two research questions from the introduction. To test whether interpretability-guided pruning outperforms iterative pruning, we run both methods at the same compression levels and compare quality directly. To test robustness to quantization, we apply GPTQ after each recovery condition and measure the quality difference. Our experimental conditions cover all combinations of pruning method, compression level, recovery regime, and language pair, giving 98 conditions in total. We chose compression levels of 8, 12, and 16 layers to match Moslem et al. [2025] and allow direct comparison. We also run a prune-only version for every configuration to see the quality impact of pruning without any recovery. For the heuristic baseline we additionally run a full fine-tuning variant to compare directly with Moslem et al. [2025]. Finally, the threshold pruning experiments are a complement to the fixed-

count ones, where instead of deciding how many layers to remove beforehand, the IFR scores decide based on how much each layer contributes to the model.

6 Results

We organize our results around the two research questions. We first look at translation quality across pruning methods and compression levels (Section 6.1), then look at quantization robustness (Section 6.2), inference speed (Section 6.3), and close with additional findings (Section 6.4).

6.1 RQ1: Does IFR-guided pruning outperform iterative pruning?

Table 2 shows the translation quality for all pruning methods and compression levels with FT + KD recovery for both language pairs. Figure 1 gives a visual summary, plotting COMET against layers kept for each method and language pair.

At 8 layers removed (24 layers kept), IFR outperforms the heuristic by 0.5 COMET points on both language pairs. At 12 layers removed the gap grows slightly, with IFR ahead by 0.9 points on Czech→German and 1.5 points on English→Spanish. At 16 layers removed the result splits: on Czech→German the heuristic wins by 2.5 points, but on English→Spanish IFR still wins by 0.9 points.

Both methods concentrate their pruning in the middle layers of the network, with IFR scores showing a clear band of low-importance layers roughly between layers 6 and 20. The Jaccard similarity between IFR and the heuristic grows from 0.14 at 8 layers removed to 0.78 at 16 layers removed on Czech→German, meaning both methods largely agree on which layers to remove at aggressive compression. The quality difference at 16 layers on Czech→German is therefore not about which layers are removed, but about the fact that the heuristic directly optimizes for chrF++ at each step while IFR scores are computed once before any pruning.

Beyond quality, IFR is significantly cheaper to compute. Scoring all 32 layers takes a single forward pass over 200 examples, roughly 15 minutes on our hardware. The iterative heuristic requires around 392 held-out evaluations at the 16-layer target, taking 2–3 hours per configuration.

The IFR threshold variants add an interesting data point. At $\lambda=0.3$ only 2 layers are removed and quality is essentially the same as the unpruned base-

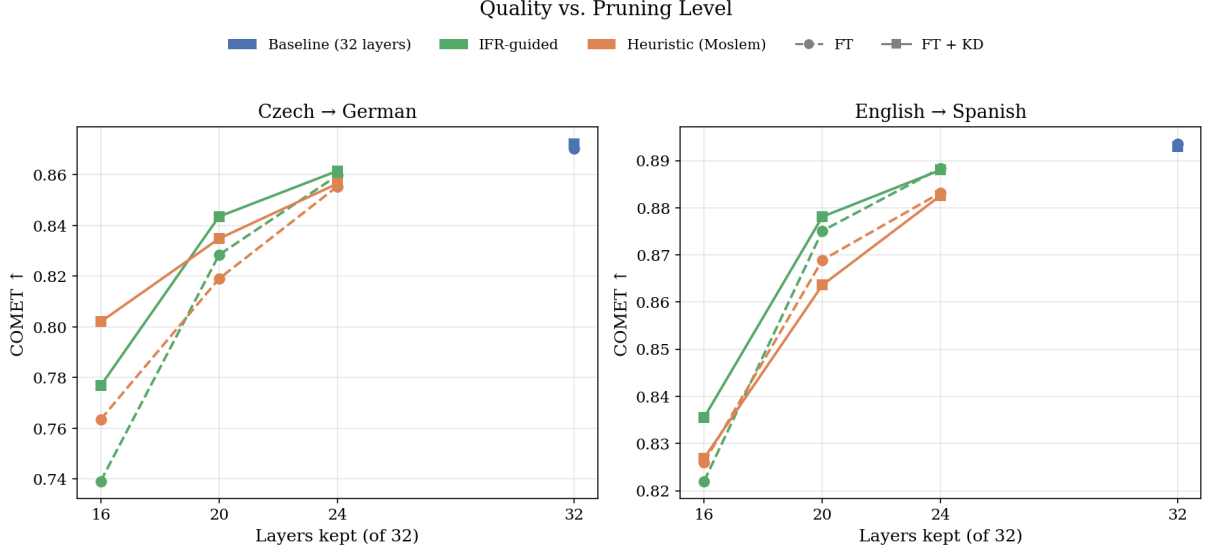


Figure 1: **Quality vs. pruning level.** COMET scores for IFR-guided and heuristic pruning at each compression level, for Czech→German (left) and English→Spanish (right). Solid lines show FT + KD recovery; dashed lines show FT only. The blue marker at 32 layers is the unpruned baseline. IFR consistently meets or exceeds the heuristic at 24 and 20 layers, with the gap reversing only at 16 layers on Czech→German.

Table 2: **RQ1 — Quality vs. compression (FT + KD recovery).** IFR-guided pruning versus heuristic iterative pruning at matched layer counts. The last two IFR rows report threshold-based variants ($t=0.3$ and $t=0.5$), which auto-select the number of layers to prune from the IFR score distribution. **Bold** marks the best result in each metric column; underline marks the runner-up. COMET $\times 100$.

Method	Layers	CES-DEU			ENG-SPA		
		chrF++↑	COMET↑	BLEU↑	chrF++↑	COMET↑	BLEU↑
Baseline	32	52.93	87.23	23.83	<u>68.08</u>	<u>89.30</u>	<u>47.35</u>
IFR	24	51.38	86.15	21.85	67.13	88.80	46.02
	20	49.96	84.35	21.08	64.97	87.81	43.13
	16	46.10	77.68	16.10	60.64	83.55	37.55
	$t=0.3$ (30)	<u>52.59</u>	<u>87.00</u>	<u>23.51</u>	68.25	89.29	47.59
	$t=0.5$ (22/23)	50.67	85.54	21.13	66.85	88.63	45.47
Heuristic	24	50.87	85.65	21.56	66.65	88.26	45.51
	20	48.95	83.48	19.26	63.70	86.36	40.70
	16	47.08	80.20	17.35	59.11	82.69	35.08

line. At $\lambda=0.5$ around 10 layers are removed automatically and quality reaches 85.54 COMET on Czech→German and 88.63 on English→Spanish, which is actually better than the fixed-count IFR model at 20 layers (84.35 and 87.81). This suggests that letting the score distribution decide the pruning depth can be better than fixing a target count.

6.2 RQ2: Are the recovered models robust to quantization?

Table 3 shows the COMET drop from fp16 to GPTQ-4bit for all configurations. Across almost every case the drop stays below 1 COMET point. The baseline loses only 0.02 points on

Czech→German and 0.06 on English→Spanish. IFR and heuristic models at 24 layers lose between 0.07 and 0.31 points. The only cases where the drop is larger are the most aggressively pruned heuristic models at 16 layers, which lose 1.87 and 1.81 points respectively. This is a clean positive answer to our second research question: pruned and recovered models are robust to GPTQ-4bit quantization, and the quality loss is negligible at moderate compression levels.

Table 3: **RQ2 — Quantization robustness (FT + KD checkpoints)**. COMET before and after GPTQ-4bit quantization. Δ columns report the absolute drop from fp16 to GPTQ-4bit (smaller is better). COMET $\times 100$.

Method	Layers	CES-DEU			ENG-SPA		
		COMET _{fp16}	COMET _{GPTQ}	Δ	COMET _{fp16}	COMET _{GPTQ}	Δ
Baseline	32	87.23	87.21	0.02	89.30	89.24	0.06
IFR	24	86.15	86.08	0.07	88.80	<u>88.56</u>	<u>0.24</u>
	20	84.35	83.87	0.48	87.81	—	—
	16	77.68	76.65	1.03	83.55	83.28	0.27
Heuristic	24	85.65	85.39	0.26	88.26	87.95	0.31
	20	83.48	82.66	0.82	86.36	85.87	0.49
	16	80.20	78.33	1.87	82.69	80.88	1.81

6.3 Inference Speed

Table 4: **Model cost and inference speed per architecture**. Parameter count, on-disk size, and vLLM throughput averaged over all FT and FT + KD checkpoints at each layer count, measured on a single A100 80 GB. **Bold** marks the best value in each column; underline marks the runner-up.

L	Params (B) $\downarrow$	fp16 (GB) $\downarrow$	GPTQ4 (GB) $\downarrow$	tok/s fp16 $\uparrow$	tok/s GPTQ $\uparrow$	Speedup $\uparrow$
32	8.03	15.0	5.4	1308	2024	1.55 $\times$
24	6.28	11.7	4.5	1513	2279	<u>1.51</u> $\times$
20	<u>5.41</u>	<u>10.1</u>	<u>4.1</u>	<u>1749</u>	<u>2570</u>	<u>1.47</u> $\times$
16	4.54	8.5	3.7	1993	2671	1.34 $\times$

Table 4 shows that pruning and GPTQ compose well. Every 8 layers removed adds roughly 200 tokens per second at fp16. GPTQ adds another 500–700 tokens per second on top of that. The 16-layer GPTQ model reaches 2671 tokens per second compared to 1308 for the fp16 baseline, more than $2\times$ faster at one third the size. Looking at the quality–speed trade-off across all configurations, IFR models with FT + KD and GPTQ consistently offer the best quality at any given inference speed on both language pairs. The best deployment point on Czech→German is the IFR 24-layer FT + KD + GPTQ model at 86.08 COMET, 4.5 GB on disk, and 2279 tokens per second. On English→Spanish it is the IFR threshold $t=0.5$ FT + KD + GPTQ model at 88.48 COMET, 4.4 GB, and 2878 tokens per second.

6.4 Additional Results

We briefly report three additional findings. First, the difference between FT and FT + KD recovery is small at moderate compression (less than 0.2 COMET points at 24 layers for both methods) but grows at aggressive pruning on Czech→German, where KD adds up to 3.8 points at 16 layers, justifying its use in our main experiments. Second, the

IFR threshold variant at $\lambda=0.5$ removes around 10 layers automatically and reaches 85.54 COMET on Czech→German and 88.63 on English→Spanish, slightly better than the fixed-count IFR model at 20 layers, suggesting that adaptive pruning depth can outperform fixed targets. Finally, BitsAndBytes [Dettmers et al., 2022, 2023] quantization is not a viable deployment option: BnB INT4 runs at 36 to 63 tokens per second with HuggingFace, while GPTQ with vLLM runs at over 2000 tokens per second, a speedup of 28 to 56 times.

7 Conclusion

In this work we explored interpretability-guided layer pruning as an alternative to iterative heuristic pruning for compressing multilingual LLMs for machine translation. We used IFR to identify the least important layers before pruning, followed by LoRA fine-tuning and Knowledge Distillation for recovery, and GPTQ quantization for deployment.

To answer our first research question: IFR-guided pruning partially outperforms the iterative baseline. At 24 and 20 layers kept, IFR matches or beats the heuristic on both language pairs. At 16 layers, the iterative method does better on Czech→German, though IFR still wins on English→Spanish. To answer our second research question: yes, the recovered models are robust to GPTQ quantization, with quality drops below 1 COMET point in almost every case.

Beyond the quality comparison, the main practical advantage of IFR is its simplicity. Running iterative pruning on Aya Expanse 8B to a 16-layer target requires around 392 held-out evaluations of 200 sentences each. IFR requires a single forward pass over 200 examples. Both methods suffer catastrophic quality loss without a recovery step, which means that a good fine-tuning strategy is just as important as the pruning method itself. GPTQ

quantization on top of the recovered models is the biggest win with the least effort, producing models that are one third the size and more than $2\times$ faster with minimal quality loss.

8 Future Work

There are several directions we would like to explore in future work. First, we only tested GPTQ 4-bit quantization as a post-recovery step. Other quantization methods such as AWQ [Lin et al., 2024] or more recent approaches could offer better quality-size trade-offs and are worth exploring.

Second, the IFR-guided method consistently struggles at aggressive compression on Czech→German compared to the iterative baseline. This raises an interesting question about why IFR scores do not fully capture the importance of the latter layers in this setting. Exploring other interpretability-guided methods such as attribution-guided pruning [Hatefi et al., 2025], or understanding what the iterative search is capturing that IFR misses, could lead to better principled pruning strategies.

Finally, both methods require a full fine-tuning recovery step after pruning, which adds significant compute. It is worth investigating whether more surgical recovery approaches could work, for example targeting only the layers adjacent to the pruned ones. Whether the quality degradation after pruning is fundamentally an architectural problem or something that can be addressed with a faster, more targeted intervention remains an open question.

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